

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

Aim: To understand the process of convolution over the image and apply over the classification problem

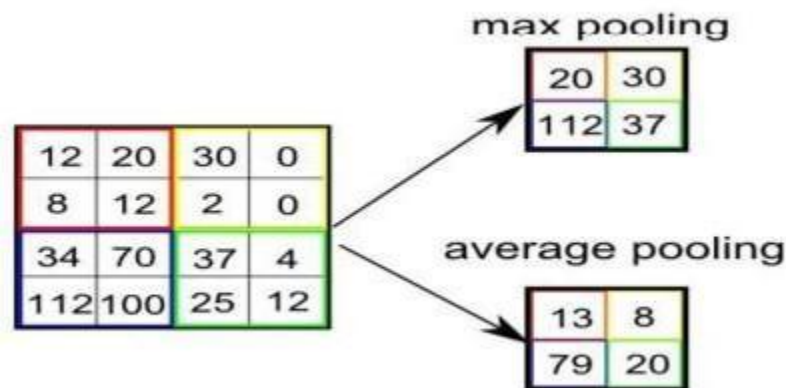
IDE: Google Colab

Theory:

Convolutional Neural Networks (CNN) are complex feed forward neural networks. CNNs are used for image classification and recognition because of its high accuracy. There are three types of layers in a convolutional neural network: i. Convolutional layer ii. Pooling layer iii. Fully connected layer. Each of these layers has different parameters that can be optimized and performs a different task on the input data.

What is Pooling Layer?

Pooling layer is responsible for reducing the spatial size of the Convolved Feature. This is to decrease the computational power required to process the data through dimensionality reduction. There are two types of Pooling: i. Average Pooling. ii. Max Pooling. Max Pooling returns the maximum value from the portion of the image covered by the Kernel. On the other hand, Average Pooling returns the average of all the values from the portion of the image covered by the Kernel. Max Pooling also performs as a Noise Suppressant. It discards the noisy activations altogether and also performs de-noising along with dimensionality reduction. On the other hand, Average Pooling simply performs dimensionality reduction as a noise suppressing mechanism. Hence, we can say that Max Pooling performs a lot better than Average Pooling.



What is Convolutional Layer?

Convolutional layers are the major building blocks used in convolutional neural networks. A convolution is the simple application of a filter to an input that results in an activation. Repeated application of the same filter to an input results in a map of activations called a feature map, indicating the locations and strength of a detected feature in an input, such as an image. A convolutional layer contains a set of filters whose parameters need to be learned. The height and weight of the filters are smaller than those of the input volume. Each filter is convolved with the input volume to compute an activation map made of neurons.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

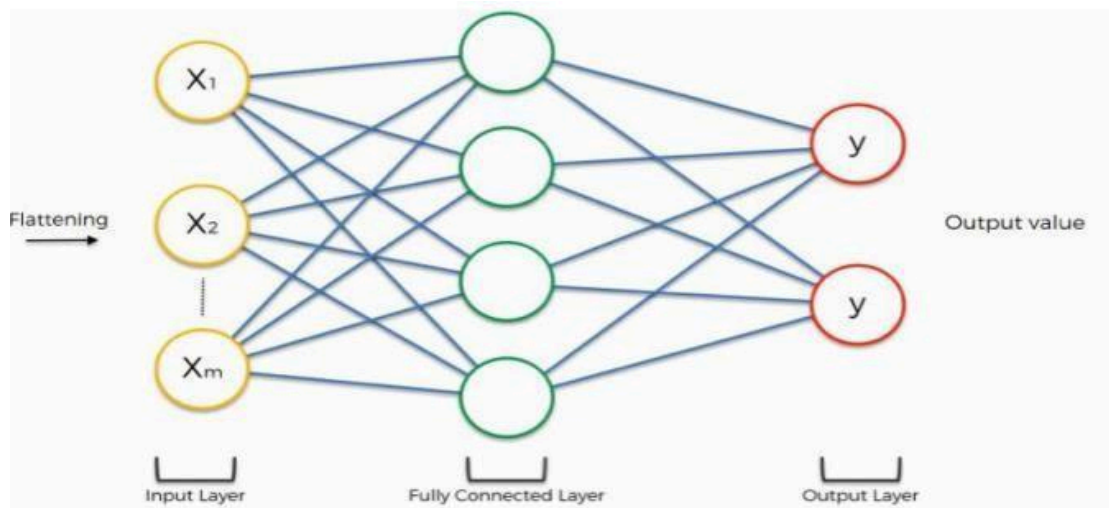
What is Fully Connected Layer?

A fully connected layer that takes the output of convolution/pooling and predicts the best label to describe the image. We have three layers in the full connection step i. Input layer ii. Fully-connected layer iii. Output layer

Input Layer: It takes the output of the previous layers, “flattens” them and turns them into a single vector that can be an input for the next stage.

Fully Connected Layer: It takes the inputs from the feature analysis and applies weights to predict the correct label.

Output Layer: It gives the final probabilities for each label.



ReLU Layer: ReLU is an activation function. Rectified Linear Unit (ReLU) transform function only activates a node if the input is above a certain quantity, while the input is below zero, the output is zero, but when the input rises above a certain threshold, it has a linear relationship with the dependent variable. The main aim is to remove all the negative values from the convolution. All the positive values remain the same but all the negative values get changed to zero.

Pre Lab Exercise:

- What is Convolution process? Explain giving example.

Convolution is a process where a **filter (kernel)** is applied to an input image to extract important features. The filter moves over the image and performs multiplication and addition to produce a **feature map**.

Example:

If we apply an edge-detection filter on an image, it highlights edges and boundaries of objects.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

b. What are different layers in CNN model?

CNN mainly consists of three layers:

- **Convolutional Layer** → extracts features from input
- **Pooling Layer** → reduces size of data
- **Fully Connected Layer** → performs final classification

c. What is the requirement of the pooling layer in the CNN model?

- Pooling layer is used to **reduce the size of feature maps**, which decreases computation.
- It also helps in **removing noise** and improving efficiency.
- Example: Max pooling selects the maximum value from a region.

d. What is the requirement of the use of ReLU activation function after convolution step?

- ReLU is used to **introduce non-linearity** in the model.
- It converts all negative values to zero and keeps positive values unchanged.
- This helps in faster training and better performance.

Methodology:

1. Load the basic libraries and packages

```
[32]
✓ 9s
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
```

2. Load the dataset

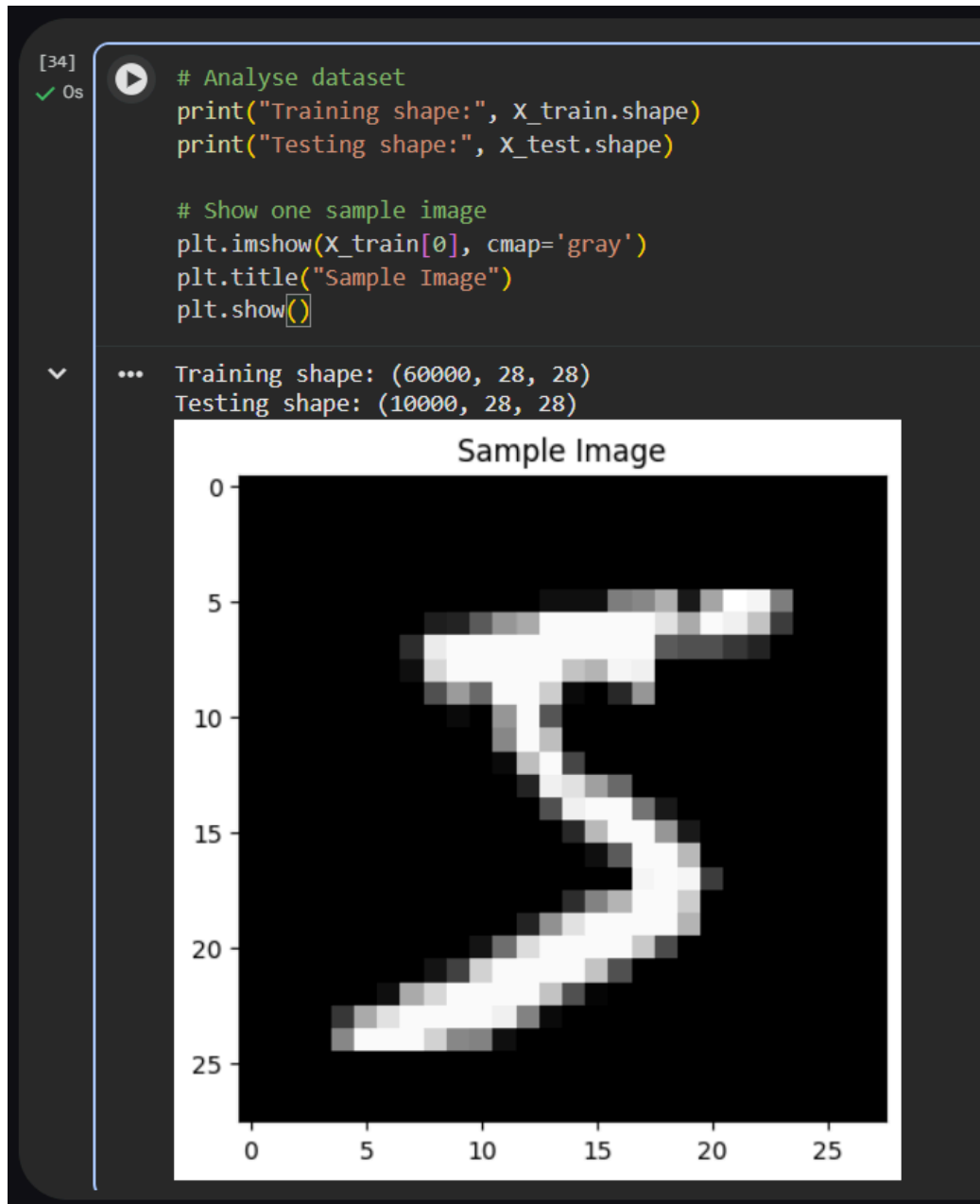
```
[33]
✓ 0s
# Load dataset
(X_train, y_train), (X_test, y_test) = keras.datasets.mnist.load_data()

X_train.shape, X_test.shape|

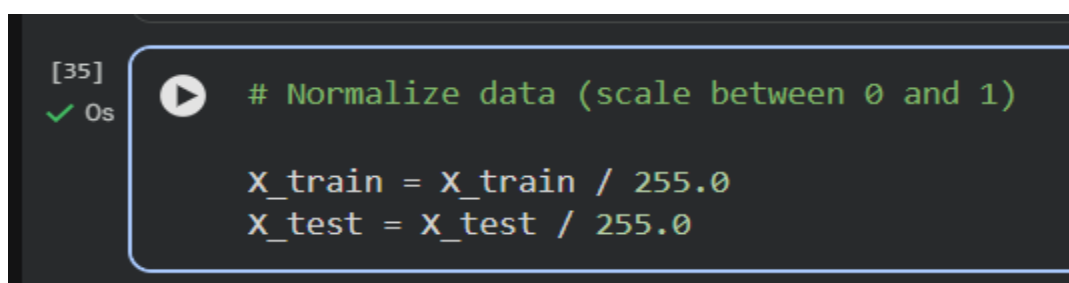
... Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 — 0s 0us/step
((60000, 28, 28), (10000, 28, 28))
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

3. Analyse the dataset



4. Normalize the data



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

5. Pre-process the data

```
[36]
✓ 0s # Reshape data for CNN (add channel dimension)

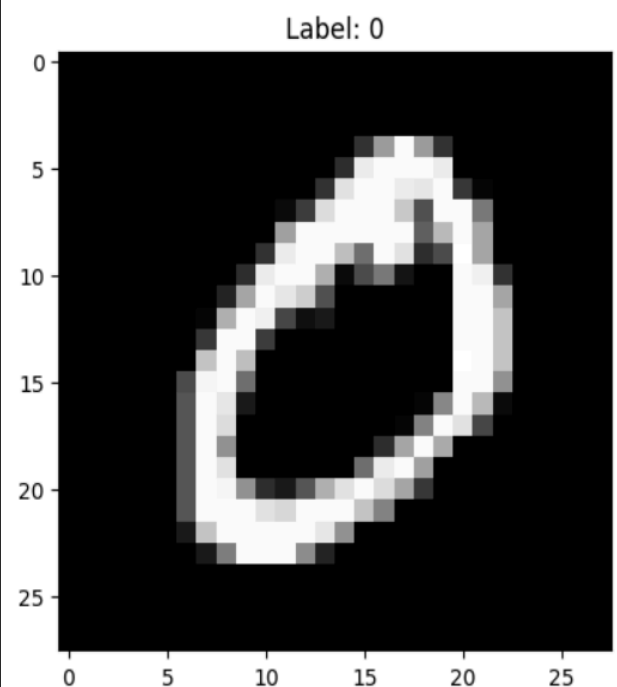
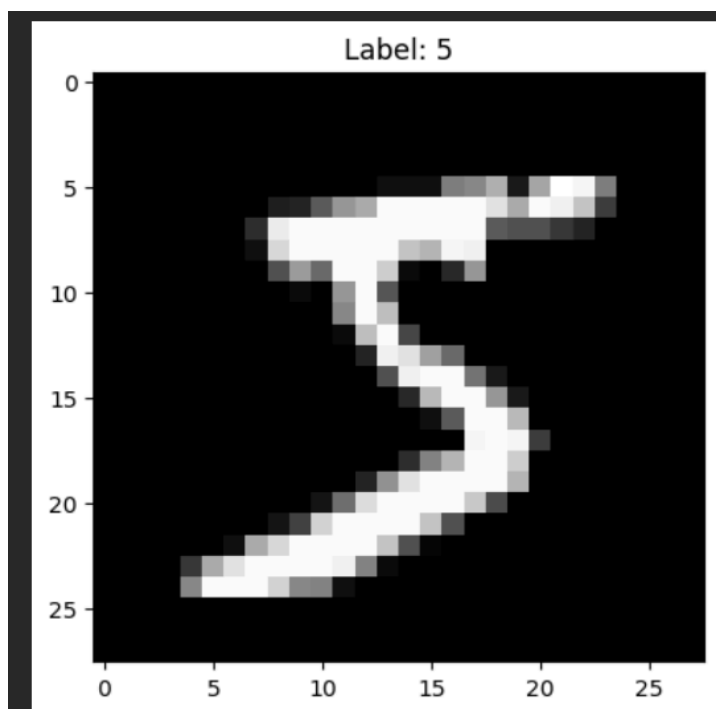
X_train = X_train.reshape(-1, 28, 28, 1)
X_test = X_test.reshape(-1, 28, 28, 1)

X_train.shape
... (60000, 28, 28, 1)
```

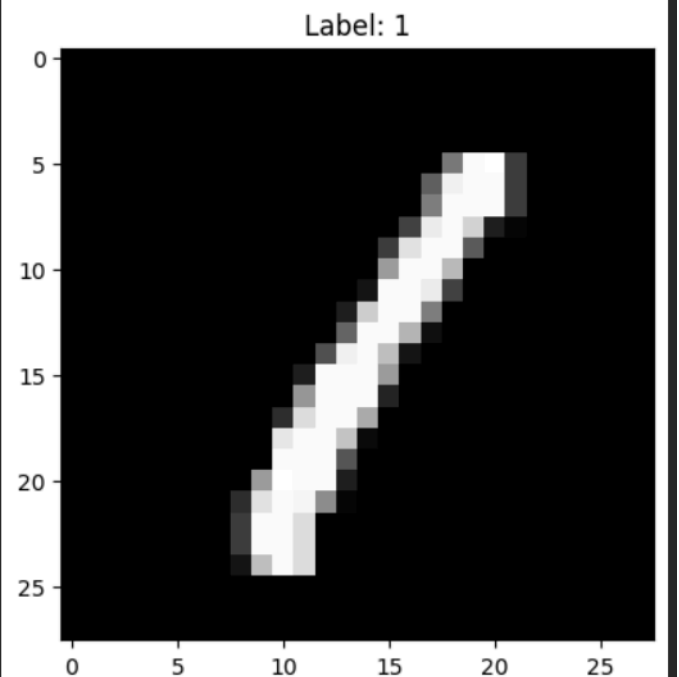
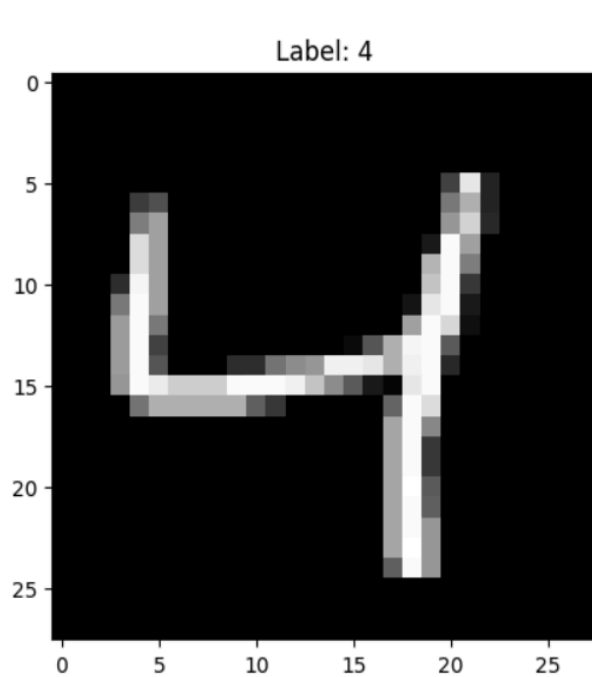
6. Visualize the Data

```
[37]
✓ 0s # Show few images

for i in range(5):
    plt.imshow(X_train[i].reshape(28,28), cmap='gray')
    plt.title(f"Label: {y_train[i]}")
    plt.show()
```



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034



7. Write the CNN model function

```
[38]
✓ 0s # Create CNN model

model = keras.Sequential([

    layers.Conv2D(32, (3,3), activation='relu', input_shape=(28,28,1)),
    layers.MaxPooling2D((2,2)),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),

    layers.Flatten(),

    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.summary()
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

```
... /usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113:
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	320
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten (Flatten)	(None, 1600)	0
dense (Dense)	(None, 64)	102,464
dense_1 (Dense)	(None, 10)	650

```
Total params: 121,930 (476.29 KB)
Trainable params: 121,930 (476.29 KB)
Non-trainable params: 0 (0.00 B)
```

8. Write the Cost Function

```
[39]
✓ Os # Compile model (define cost function + optimizer)

model.compile([
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
])
```

9. Write the Gradient Descent optimization algorithm

```
[40]
✓ 5m # Train model

history = model.fit(
    x_train, y_train,
    epochs=5,
    validation_data=(x_test, y_test)
)
```

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

```

... Epoch 1/5
1875/1875 ————— 56s 29ms/step - accuracy: 0.9606 - loss: 0.1312 - val_accuracy: 0.9870 - val_loss: 0.0436
Epoch 2/5
1875/1875 ————— 54s 29ms/step - accuracy: 0.9861 - loss: 0.0449 - val_accuracy: 0.9913 - val_loss: 0.0294
Epoch 3/5
1875/1875 ————— 82s 29ms/step - accuracy: 0.9909 - loss: 0.0302 - val_accuracy: 0.9914 - val_loss: 0.0256
Epoch 4/5
1875/1875 ————— 54s 29ms/step - accuracy: 0.9928 - loss: 0.0237 - val_accuracy: 0.9919 - val_loss: 0.0250
Epoch 5/5
1875/1875 ————— 54s 29ms/step - accuracy: 0.9950 - loss: 0.0167 - val_accuracy: 0.9910 - val_loss: 0.0288

```

10. Apply the training over the dataset to minimize the loss

```

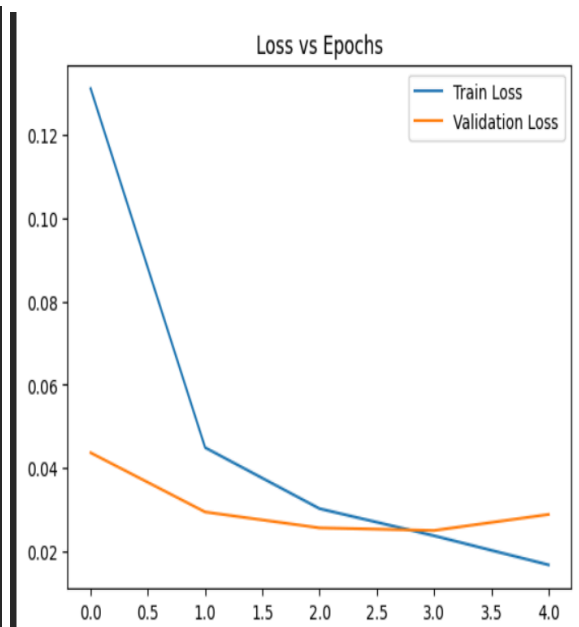
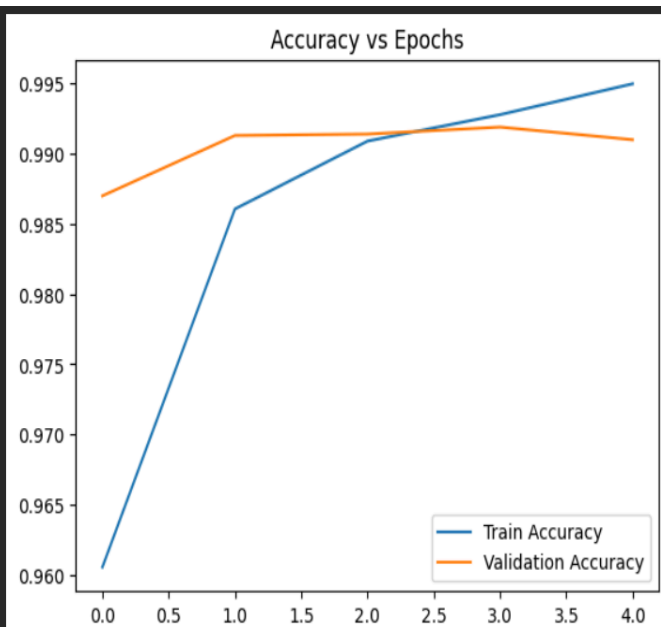
[41]
✓ Os # Plot accuracy

plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.legend()
plt.title("Accuracy vs Epochs")
plt.show()

# Plot loss

plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.legend()
plt.title("Loss vs Epochs")
plt.show()

```



 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

11. Observe the cost function vs iterations learning curve

The learning curve shows that loss decreases and accuracy increases with epochs, indicating proper training of the CNN model.

Results:

To be attached with

- a. Training dataset
MNIST dataset is used which contains handwritten digit images for training and testing.
- b. Model summary
Model consists of convolution, pooling and fully connected layers for image classification.
- c. Training and validation accuracy w.r.t epochs before regularization
Training accuracy increases with epochs, while validation accuracy also improves but may show slight variation.
- d. Training and validation loss w.r.t epochs before regularization
Training loss decreases steadily, while validation loss decreases initially and may fluctuate.
- e. Training and validation accuracy w.r.t epochs after regularization
Accuracy becomes more stable and overfitting is reduced after regularization.
- f. Training and validation loss w.r.t epochs after regularization
Loss decreases smoothly and model generalization improves.
- g. Original v/s predicted labels for correct predicted observations
Model correctly predicts most images where predicted labels match original labels.
- h. Original v/s predicted labels for incorrect predicted observations
Some predictions are incorrect due to similar patterns or noise in images.

Observation and Result Analysis:

- e. Nature of the dataset
Dataset consists of grayscale images of handwritten digits with clear patterns for classification.
- f. Training Process without regularization
Model learns patterns and accuracy increases, but may show overfitting with higher training accuracy than validation.
- g. After regularization in the training Process
Overfitting is reduced and validation accuracy improves, making the model more stable.
- h. Observation over the Learning Curves
Accuracy increases and loss decreases over epochs, indicating proper learning of the model.

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To understand the process of convolution over the image and apply over the classification problem	
Experiment No: 4	Date:	Enrolment No:92410133034

Post Lab Exercise:

- a. Why CNN is preferred over ANN for images

CNN is preferred because it can automatically extract spatial features like edges and patterns from images, while ANN cannot efficiently handle image data.

- b. Can CNN be applied over Text data? If yes, then how. If no, then why?

Yes, CNN can be applied on text data by converting text into numerical form (word embeddings) and then applying convolution to extract important patterns.

- c. What is the role of dropout layer?

Dropout layer randomly removes some neurons during training to prevent overfitting and improve model generalization.

- d. What will happen if maxpooling is replaced with minpooling?

in pooling will select minimum values instead of important features, which may reduce model performance and accuracy.”